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Neuromorphic foreign object detection

Abstract

A neuromorphic foreign object debris (FOD) detection system includes a FOD processing system and a neuromorphic sensor. The FOD processing system includes a FOD controller including a trained artificial intelligence machine learning (AIML) model representing an area of interest. The neuromorphic sensor has a field of view (FOV) containing the area of interest and is configured to output pixel data in response to FOD appearing in the FOV. The FOD controller detects the FOD is present in the area of interest in response to receiving the pixel data, and generates an alert signal indicating the presence of the FOD.

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Background/Summary

BACKGROUND

(1) Exemplary embodiments of the present disclosure pertain to aircraft engine production and maintenance, and more particularly, to foreign object debris detection in the presence of aircraft engines.

(2) FOD (Foreign Object Debris) is material that is present that has the potential to significantly damage or destroy an aircraft engine that is at test in a test cell or in operation on an aircraft in ground operations. FOD material can be brought in by wind, foot traffic, vehicle traffic, fall off of machines, equipment, and/or be dropped by personnel. Current solutions to prevent FOD damage typically involves manual inspection of test cell and shop floors and aircraft taxiways and runways. Although the ingestion of FOD material into the engine is uncommon, the consequences of ingestion are substantial and include engine damage and/or complete engine failure.

BRIEF DESCRIPTION

(3) According to a non-limiting embodiment, a neuromorphic foreign object debris (FOD) detection system includes a FOD processing system and a neuromorphic sensor in signal communication with the FOD processing system. The FOD processing system includes a FOD controller including a trained artificial intelligence machine learning (AIML) model representing an area of interest. The neuromorphic sensor has a field of view (FOV) containing the area of interest and is configured to output pixel data in response to FOD appearing in the FOV. The FOD controller detects the FOD is present in the area of interest in response to receiving the pixel data, and generates an alert signal indicating the presence of the FOD.

(4) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the AIML model is implemented as a deep convolutional neural network (DCNN) model.

(5) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the AIML model is implemented as a deep convolutional auto encoder (CAE) model).

(6) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor includes an event camera.

(7) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the area of interest includes an aircraft engine.

(8) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is located remotely from the aircraft engine and the FOV contains the aircraft engine.

(9) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is coupled to an aircraft engine and the FOV contains the aircraft engine.

(10) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is disposed inside the aircraft engine.

(11) According to another non-limiting embodiment, a method of performing neuromorphic foreign object debris (FOD) detection comprises generating training data and training an artificial intelligence machine learning (AIML) model using the training data depicting an area of interest. The method further comprises monitoring the area of interest using a neuromorphic sensor, and outputting pixel data from the neuromorphic sensor in response to FOD appearing in the area of interest. The method further comprises generating an alert indicating a presence of FOD in the area of interest in response based on the pixel data.

(12) In addition to one or more of the features described above, or as an alternative to any of the

foregoing embodiments, the AIML model is implemented as a deep convolutional neural network (DCNN) model.
(13) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the AIML model is implemented as a deep convolutional auto encoder (CAE) model.
(14) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the area of interest includes an aircraft engine.
(15) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is located remotely from the aircraft engine and the FOV contains the aircraft engine.
(16) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is coupled to an aircraft engine and the FOV contains the aircraft engine.
(17) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is disposed inside the aircraft engine.
(18) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the AIML model is implemented as a deep convolutional neural network (DCNN) model.
(19) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the AIML model is implemented as a deep convolutional auto encoder (CAE) model.
(20) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the area of interest includes an aircraft engine.
(21) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is located remotely from the aircraft engine and the FOV contains the aircraft engine.
(22) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is coupled to an aircraft engine and the FOV contains the aircraft engine.
(23) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the neuromorphic sensor is disposed inside the aircraft engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
(2) FIG. **1** depicts a neuromorphic FOD detection system according to a non-limiting embodiment of the present disclosure; and
(3) FIG. **2**A illustrates a supervised training process to train an artificial intelligence machine learning (AIML) algorithm/model employed the neuromorphic FOD detection system according to a non-limiting embodiment of the present disclosure;
(4) FIG. **2**B illustrates an unsupervised training process to train an artificial intelligence machine learning (AIML) algorithm/model employed the neuromorphic FOD detection system according to non-limiting embodiment of the present disclosure;
(5) FIG. **3**A depicts a neuromorphic sensor capturing an image according to a non-limiting embodiment of the present disclosure;
(6) FIG. **3**B depicts a field of view (FOV) of the neuromorphic sensor shown in FIG. **3**A according to a non-limiting embodiment of the present disclosure;

(7) FIG. **4**A depicts the neuromorphic sensor of FIG. **3**A detecting FOD in the image according to another non-limiting embodiment of the present disclosure;
(8) FIG. **4**B depicts the FOV of FIG. **3**B including FOD according to another non-limiting embodiment of the present disclosure;
(9) FIG. **5**A depicts a neuromorphic sensor capturing an image according to another non-limiting embodiment of the present disclosure;
(10) FIG. **5**B depicts a field of view (FOV) of neuromorphic sensor shown in FIG. **5**A according to another non-limiting embodiment of the present disclosure;
(11) FIG. **6**A depicts the neuromorphic sensor of FIG. **5**A detecting FOD in the image according to another non-limiting embodiment of the present disclosure;
(12) FIG. **6**B depicts the FOV of FIG. **5**B including FOD according to another non-limiting embodiment of the present disclosure; and
(13) FIG. **7** is a flow diagram illustrating a method of performing neuromorphic FOD detection according to another non-limiting embodiment of the present disclosure.

DETAILED DESCRIPTION

(14) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(15) Traditional cameras (sometimes referred as “RGB pixel cameras” or “shutter cameras”) capture entire images in the camera's field of view (FOV) each time the shutter is opened. As a result, a large amount of pixel data is captured and in turn requires a large amount of memory and processing power to process the captured pixel data. Moreover, high-speed imaging necessary for object detection, for example, has proven to be impractical using traditional cameras due to the large amounts of memory space required for storage of high speed videos, and the intensive time-consuming task to view them frame by frame.

(16) In contrast to traditional cameras (e.g., RGB pixel cameras or shutter cameras), a neuromorphic sensor (sometimes referred to as an “event camera”) is an imaging sensor that responds to local changes in brightness instead of capturing the full image in the FOV using a camera shutter. Each pixel in a pixel array of the neuromorphic sensor operates independently and asynchronously, each reporting changes in brightness as they occur (referred to herein as an “event”). For example, each pixel stores a reference brightness level (e.g., a preset threshold), and continuously compares the reference brightness level to a current level of brightness. If a difference in brightness exceeds the reference brightness level, the pixel resets the reference brightness level and generates an indication of the event, which can comprise a data packet of information or message containing the pixel's address (e.g., x, y, or other spatial coordinates in the pixel array), a timestamp indicating a time of the event (i.e., a time that the event occurred), and a value representing a change in brightness detected by the pixel (e.g., a polarity (increase or decrease) of a brightness change, or a measurement of a current level of illumination). Accordingly, the neuromorphic sensor can provide a focal plane array (FPA) sensitivity, dynamic range, and an effective frame rate to enable Fourier Transform spectroscopy for scenes in motion.

(17) Various embodiments of the present disclosure provides a neuromorphic foreign object debris (FOD) detection system, which implements one or more neuromorphic sensors configured to detect FOD in an area of interest such as, for example, near an aircraft engine. The FOD system can employ an artificial intelligence machine learning (AIML) algorithm or model such as a deep convolutional neural network (DCNN) model and/or a deep convolutional auto encoder (CAE) model, which is trained using image and/or video data depicting a FOV or scene of an area of interest in which to detect FOD. As descried herein, the neuromorphic sensor outputs only the pixel data associated with the pixels containing the changes in the image. Accordingly, the imaging system is able to process a lower amount of pixel data and generate an alert indicating a change in the image scene such as, for example, when FOD is located in the presence of an aircraft engine. In one or more non-limiting embodiments, the neuromorphic FOD detection system is capable of

detecting the presence of FOD and automatically generating an alert of the presence of FOD. In this manner, a service technician can be dispatched to remove the FOD and prevent it from being ingested into the engine.

(18) With reference now to the drawings, FIG. **1** depicts a neuromorphic foreign object debris (FOD) detection system **100** that includes a processing system **102** and one or more neuromorphic sensors **104**. The neuromorphic sensors **104** can include one or more “event cameras” **104**, which can analyze one or more structures **108** and objects within a FOV **110**. The structure **108** described herein is an aircraft engine such as gas turbine engine **108**, for example, but it should appreciated that the neuromorphic sensors **104** descried herein can analyze other types of structures **108** without departing from the scope of the invention.

(19) The processing system **102** includes at least one processor **114**, memory **116**, and a sensor interface **118**. The processing system **102** can also include a user input interface **120**, a display interface **122**, a network interface **124**, and other features known in the art. The neuromorphic sensors **104** are in signal communication with the sensor interface **118** via wired and/or wireless communication. In this manner, pixel data output from the neuromorphic sensors **104** can be delivered to the processing system **102** for processing.

(20) The processor **114** can be any type of central processing unit (CPU), or graphics processing unit (GPU) including a microprocessor, a digital signal processor (DSP), a microcontroller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or the like. Also, in embodiments, the memory **116** may include random access memory (RAM), read only memory (ROM), or other electronic, optical, magnetic, or any other computer readable medium onto which is stored data and algorithms as executable instructions in a non-transitory form.

(21) The processor **114** and/or display interface **122** can include one or more graphics processing units (GPUs) which may support vector processing using a single instruction multiple data path (SIMD) architecture to process multiple layers of data substantially in parallel for output on display **126**. The user input interface **120** can acquire user input from one or more user input devices **128**, such as keys, buttons, scroll wheels, touchpad, mouse input, and the like. In some embodiments the user input device **128** is integrated with the display **126**, such as a touch screen. The network interface **124** can provide wireless and/or wired communication with one or more remote processing and/or data resources, such as cloud computing resources **130**. The cloud computing resources **130** can perform portions of the processing described herein and may support model training.

(22) FIG. **2**A illustrates a training process **200** using supervised learning **202** to train an artificial intelligence machine learning (AIML) algorithm or model **204** executed by the neuromorphic FOD detection system **100** according to a non-limiting embodiment of the present disclosure. In the example of FIG. **2**A, a data source **206** provides training data **205** to develop the AIML algorithm or model **204**, which when developed represent an area of interest in which to detect FOD. The training data **205** depicts an area of interest in which to FOD is to be detected. In the example illustrated in FIG. **2**A, the training data **205** is used to develop the AIML algorithm or model **204** after preprocessing **208** is performed. It should be appreciated, however, that the preprocessing **208** can be omitted and the data source **206** can provide the data **205** directly to the AIML algorithm or model **204** without departing from the scope of the invention.

(23) The training data **205** in data source **206** can originate from data captured by one or more of the neuromorphic sensors **104** shown in FIG. **1** during a training phase. The training process **200** may be performed as part of an off-line process using a separate processing system other than the processing system **102** of FIG. **1**. Alternatively, the processing system **102** may be configured in a training phase to implement the training process **200** of FIG. **2**A.

(24) In the example of FIG. **2**A, the area of interest depicted by the training data **205** includes video/image data (e.g., images **207***a*, **207***b*, **207***n*) depicting an aircraft gas turbine engine **108** installed on a testing rig captured according to different views and orientations. It should be

appreciated that images of other areas of interest (e.g. other than an aircraft engine) can be used as training data to train the neuromorphic FOD detection system to detect foreign objects in different environments. For example, one or more neuromorphic sensors **104** can be coupled to the aircraft itself (e.g., to a rear portion of the aircraft or inside the engine) and orientated with a FOV **110** that includes the aircraft engine **108**. The neuromorphic FOD detection system **100** can then trained to identify different views of the taxiway or runway to detect FOD that may come in close proximity to the aircraft engine.

(25) For purposes of training, images **207***a*, **207***b*, **207***n* of a gas turbine engine **108** installed on a testing rig appearing in different views are labeled as such are used to train the AIML algorithm or model **204**. Video frame data **210** from training data **205** can be provided to a region-of-interest detector **212** that may perform edge detection or other types of region detection known in the art as part of preprocessing **208**. A patch detector **214** can detect patches (i.e., areas) of interest based on the regions of interest identified by the region-of-interest detector **212** as part of preprocessing **208**. For example, a threshold can be applied on a percentage of pixels with edges in a given patch. A labeler **216** extracts label data **218** from the training data **205** and applies labels to video data **210** from selected patches of interest as detected by the patch detector **214** as part of preprocessing **208**, where labeling can be on a patch or pixel basis.

(26) For each selected patch, the labeler **216** applies the label data **218** to the frame data **210** on multiple channels. If the training data **205** includes two different labels, then the labeler **216** can apply at least one new label normal/undamaged edges). The labeled data from the labeler **216** is used for supervised learning **202** to train the AIML algorithm or model **204** using a convolutional neural network (CNN) which may also be referred to as a deep CNN or DCNN. Supervised learning **202** can compare classification results of the AIML algorithm or model **204** to a ground truth and can continue running iterations of the AIML algorithm or model **204** until a desired level of classification confidence is achieved. In this manner, the neuromorphic FOD detection system **100** can be trained to learn the “nominal” surroundings or environment of the engine **108**, i.e., when no FOD is present near the engine **108** and detect when the surroundings or environment change once FOD becomes present.

(27) According to another non-limiting embodiment, the training process **200** can use unsupervised learning to train an artificial intelligence machine learning (AIML) algorithm or model executed by the neuromorphic FOD detection system **100**. Accordingly, the neuromorphic FOD detection system **100** can learn from normal observations and detect anomalous signatures as FOD. FIG. **2**B, for example, illustrates a training process **200** using unsupervised learning **202** to train an artificial intelligence machine learning (AIML) model **204** according to an embodiment. In the example of FIG. **2**B, a data source **206** provides training data **205** to develop the artificial intelligence machine learning (AIML) model **204** after preprocessing **208** is performed. It should be appreciated, however, that the preprocessing **208** can be omitted and the data source **206** can provide the data **205** directly to the auto-encoder **204** without departing from the scope of the invention.

(28) The training data **205** in data source **206** can originate from data captured by the neuromorphic foreign object debris (FOD) detection system **100** of FIG. **1** during a training phase. The training process **200** can be performed as part of an off-line process using a separate processing system other than the processing system **102** of FIG. **1**. Alternatively, the processing system **102** can be configured in a training phase to implement the training process **200** of FIG. **2**B.

(29) In the example of FIG. **2**B, the training data **205** includes video and/or image data **207***a*, **207***b*, **207***n*, from one or more sensors, e.g., sensor **104** of FIG. **1**. Given multi-modal sensor data with no prior knowledge of the system, it is possible to register the data streams. For illustration, training process **200** is described with respect to an image-video registration example. By creating an over-constrained deep auto-encoder (DAC) definition, the DAC can be driven to capture mutual information in both the image and/or video data by reducing the randomness of a DAC bottleneck layer (i.e., a reduction layer) well beyond the rank at which optimal reconstruction occurs.

Minimizing the reconstruction error with respect to relative shifts of the image and/or video data reflects that the current alignment of the sensor data has the greatest correlation possible (i.e., smallest misalignment). This method can be applied for both spatial and temporal registration.
(30) A deep neural network auto-encoder (DNN-AE) takes an input $x \in Rd$ and first maps it to the latent representation $h \in Rd'$ using a deterministic function of the type $h=f\theta=\sigma(Wx+b)$ with $\theta=\{W, b\}$ where W is the weight and b is the bias. This "code" is then used to reconstruct the input by a reverse mapping of $y=f\theta'(h)=\sigma(W'h+b')$ with $\theta'=\{W',b'\}$. The two parameter sets are usually constrained to be of the form $W'=WT$, using the same weights for encoding the input and decoding the latent representation. Each training pattern xi is then mapped onto its code hi and its reconstruction yi. The parameters are optimized, minimizing an appropriate cost function over the training set $Dn=\{(x0, t0), \ldots (xn, tn)\}$.
(31) The first step includes using a probabilistic Restricted Boltzmann Machine (RBM) approach, trying to reconstruct noisy inputs. The training process **200** can involve the reconstruction of a clean sensor input from a partially destroyed/missing sensor. The sensor input (x) becomes corrupted sensor input (x) by adding a variable amount (v) of noise distributed according to the characteristics of the input data. An RBM network is trained initially with the same number of layers as envisioned in the final DNN-AE in model **204**. The parameter (v) represents the percentage of permissible corruption in the network. The model **204** is trained to de-noise the inputs by first finding the latent representation $h=f\theta(x)=\sigma(Wx+b)$ from which to reconstruct the original input $y=f\theta'(h)=\sigma(W'h+b')$.
(32) As part of preprocessing **208**, frame data **210** from training data **205** can be provided to a region-of-interest detector **212** that may perform edge detection or other types of region detection known in the art. A patch detector **214** can detect patches (i.e., areas) of interest based on the regions of interest identified by the region-of-interest detector **212** as part of preprocessing **208**. Data fuser **216** can merge image data **218** from the training data **205** with image and/or video data **210** from selected patches of interest as detected by the patch detector **214** as part of preprocessing **208**. The frame data **210** and image data **218** fused as multiple channels for each misalignment are provided for unsupervised learning **202** of model **204**. Although depicted as a deep convolutional auto-encoder (CAE), the model **204** can use a CAE or a DNN-AE, and more generally, a deep auto-encoder.
(33) Although not illustrated, the training process **200** can implement an autoencoder trained according to semi-supervised learning, which involves using only a few known rare FOD cases and other unlabeled FODs. The semi-supervised learning utilizes a machine learning (ML) model that can be trained according to the following operations: (1) If a label exists, directly optimize the ML model by the supervised loss (e.g., "where is the FOD? Is the predicted FOD type correct?", etc.); and (2) If label does not exist, optimize the ML model by reconstruction error. For example, the autoencoder can be trained to learn the nominal observations and reject outliers and/or anomalies, which are likely FODs of interest.
(34) Turning now to FIG. **3**A, a neuromorphic sensor **104** is illustrated capturing an image of a gas turbine engine **108** installed on a testing rig according to a non-limiting embodiment of the present disclosure. In this example, the neuromorphic sensor **104** captures the gas turbine engine **108** in the FOV **110** while existing in a nominal environment, i.e., when no FOD **304** is present near the engine **108**. FIG. **3**B depicts FOV **110** of neuromorphic sensor **104** that includes the engine **108**. The image generated by the neuromorphic sensor **104** includes a pixel array **300** defined by a plurality of individual pixels **302**.
(35) In FIGS. **4**A and **4**B, the neuromorphic sensor **104** is depicted detecting FOD **304** in the FOV **110** and in the presence of the engine **108**. The introduction of the FOD **304** changes the brightness of one or more localized pixels **303** containing the FOD **304** (as indicated by highlighted pixels **303**) without changing the brightness of pixels excluding the FOD **304**. Accordingly, the neuromorphic sensor **104** outputs pixel data indicating the affected localized pixels **303** and the

FOD **304** without outputting pixel data corresponding to the remaining pixels **302**, i.e., the pixels **302** that exclude the FOD **304**. Accordingly, the processing system **102** can detect FOD **304** is present in the surrounding environment of the engine **108** which is not normally present in the nominal environment of the engine **108**. In at least one non-limiting embodiment, the processing system **102** can automatically generate an alert (e.g., an audio alert, visual alert, haptic alert, etc.) that alerts a technician of the FOD **304** in response to detecting the FOD **304**. In this manner, the FOD **304** can be removed from the surrounding environment so that it is not ingested into the engine **108**.

(36) Turning to FIGS. **5**A and **5**B, a neuromorphic FOD detection system **100** is illustrated according to another non-limiting embodiment. In this example, the example, neuromorphic sensors **104** are coupled to an aircraft **150** and orientated to establish a FOV **110** that contains the aircraft engine **108**. As shown in FIG. **5**B, the neuromorphic sensors **104** capture the engine **108** in the FOV **110** while existing in a nominal environment, i.e., when no FOD is present near the engine **108**.

(37) In FIGS. **6**A and **6**B, FOD **304** appears in the FOV **110** (e.g., on the taxiway) and is present near the engine **108**. As described herein, the introduction of the FOD **304** changes the brightness of one or more localized pixels **303** containing the FOD **304** (as indicated by highlighted pixels **303**) without changing the brightness of pixels excluding the FOD **304**. Accordingly, one or more of the neuromorphic sensors **104** output pixel data indicating the affected localized pixels **303** and the FOD **304** without outputting pixel data corresponding to the remaining pixels **302**, i.e., the pixels **302** that exclude the FOD **304**. Accordingly, the processing system **102** can detect FOD **304** is present in the surrounding environment of the engine **108** (e.g., on the taxiway) which is not normally present in the nominal environment of the engine **108**. In response to detecting the FOD **304**, the processing system **102** can automatically trigger other sensors such as one or more high-fidelity like or speed-cameras to begin collecting data. In this manner, the amount of data associated with high-fidelity and high-speed image capturing and processing can be reduced, e.g., captured, stored and processed only when the FOD processing system **102** system identifies a FOD **304**. In addition, the FOD processing system **102** can also generate an alert (e.g., an audio alert, visual alert, haptic alert, etc.) that alerts a technician of the FOD **304**. In this manner, the FOD **304** can be removed so that it is not ingested into the engine **108**.

(38) Turning now to FIG. **7**, a method of performing neuromorphic FOD detection is illustrated according to a non-limiting embodiment. The method begins at operation **700**, and at operation **702** training data for training a neuromorphic FOD detection system is generated. At operation **704**, an AIML model utilized by the neuromorphic FOD detection system is trained using the training data. At operation **706**, a neuromorphic sensor is located near an area of interest in which to detect FOD. In one or more non-limiting embodiments, the area of interest is an aircraft engine.

(39) At operation **708**, the neuromorphic sensor detects whether FOD is present in the area of interest. When FOD is not present in the FOV of the neuromorphic sensor, the method returns to operation **706** and continues monitoring the area of interest. When, however, FOD is present in the FOV of the neuromorphic sensor, one or more pixels of the neuromorphic sensor are changed (e.g., the brightness level of the pixels change) and the neuromorphic sensor outputs changed pixel data at operation **710**. At operation **712**, the neuromorphic FOD detection system (e.g., a FOD controller) receives the changed pixel data from the neuromorphic sensor, and outputs an alert signal indicating the presence of FOD in the area of interest, e.g., near the aircraft engine. The method ends at operation **714**.

(40) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(41) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates

otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.
(42) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. Moreover, the embodiments or parts of the embodiments may be combined in whole or in part without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A neuromorphic foreign object debris (FOD) detection system comprising: a FOD processing system including a FOD controller including a trained artificial intelligence machine learning (AIML) model representing an area of interest; and a neuromorphic sensor in signal communication with the FOD processing system, the neuromorphic sensor having a field of view (FOV) containing the area of interest and configured to output pixel data in response to FOD appearing in the FOV, wherein the FOD controller detects the FOD is present in the area of interest in response to receiving the pixel data, and generates an alert signal indicating the presence of the FOD.
2. The FOD detection system of claim 1, wherein the AIML model is implemented as a deep convolutional neural network (DCNN) model.
3. The FOD detection system of claim 1, wherein the AIML model is implemented as a deep convolutional auto encoder (CAE) model.
4. The FOD detection system of claim 1, wherein the neuromorphic sensor includes an event camera including a plurality of pixels, each pixel configured to realize a local change in brightness and to output local pixel data indicating the local change in brightness as it occurs, and wherein the pixel data indicates corresponding to at least one localize pixel among the plurality of pixels that realizes the local change in brightness while excluding pixel data corresponding to remaining pixels that do not realize the local change in brightness.
5. The FOD detection system of claim 1, wherein the area of interest includes an aircraft engine.
6. The FOD detection system of claim 5, wherein the neuromorphic sensor is located remotely from the aircraft engine and the FOV contains the aircraft engine.
7. The FOD detection system of claim 5, wherein the neuromorphic sensor is coupled to an aircraft engine and the FOV contains the aircraft engine.
8. The FOD detection system of claim 5, wherein the neuromorphic sensor is disposed inside the aircraft engine.
9. A method of performing neuromorphic foreign object debris (FOD) detection comprises: generating training data; training an artificial intelligence machine learning (AIML) model using the training data depicting an area of interest; monitoring the area of interest using a neuromorphic sensor; outputting pixel data from the neuromorphic sensor in response to FOD appearing in the area of interest; and generating an alert indicating a presence of FOD in the area of interest in response based on the pixel data.
10. The method of claim 9, wherein the AIML model is implemented as a deep convolutional neural network (DCNN) model.

11. The method of claim 9, wherein the AIML model is implemented as a deep convolutional auto encoder (CAE) model.
12. The method of claim 9, wherein the area of interest includes an aircraft engine.
13. The method of claim 12, wherein the neuromorphic sensor is located remotely from the aircraft engine and the FOV contains the aircraft engine.
14. The method of claim 12, wherein the neuromorphic sensor is coupled to an aircraft engine and the FOV contains the aircraft engine.
15. The method of claim 12, wherein the neuromorphic sensor is disposed inside the aircraft engine.
16. The FOD detection system of claim 4, wherein the FOD controller detects the FOD is present in the area of interest in response to receiving the output local pixel data from at least one pixel among the plurality of pixels.
17. The method of claim 9, further comprising detecting, by at least one local pixel among a plurality of pixels included in the neuromorphic sensor, a local change in brightness independent from a change in brightness among remaining pixels included in the plurality of pixels; analyzing, by a FOC controller, the the pixel data which indicates the local change in brightness of the at least one local pixel corresponding to at least one local pixel among the plurality of pixels while excluding pixel data corresponding to the remaining pixels that do not realize the local change in brightness; and determining the presence of FOD in the area of interest based on the local change in brightness of the at least one local pixel indicated by the pixel data.